

Diversified but Crowded: Evidence from 13F Holdings

Liang Hu and Yinru Shen

Department of Computer Science, Columbia University, New York, NY, United States
Department of Mathematics, Florida State University, Tallahassee, FL, United States

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Diversified but Crowded: Evidence from 13F Holdings

Anonymous Authors

Abstract

Can an institution diversify internally while becoming more similar to its peers? Using filing-resolved SEC Form 13F holdings for the 500 largest reported portfolios each quarter from 2013Q2 through 2025Q4, we measure manager crowding as mean pairwise minimum-weight overlap. In regressions with quarter fixed effects, portfolio controls, and manager-clustered standard errors, entropy is positively associated with crowding ($\beta = 0.067$, $SE = 0.003$). A one-standard-deviation entropy difference in the estimation sample corresponds to 0.082 greater crowding, or 50.6% of the mean. Within the quarterly reselected top-500 sample, mean crowding rises from 0.129 to 0.230, while the random-breadth benchmark rises by 0.019; this aggregate trend combines portfolio changes with changes in sample composition. Security-popularity and same-quarter peer-consensus diagnostics sharply attenuate the association; exact large-manager exclusions leave it nearly unchanged. These descriptive results concern reported long positions, not complete portfolios or causal effects. They show that within-portfolio diversification and cross-institution distinctiveness are separate dimensions.

Keywords: Institutional investors, 13F filings, Portfolio crowding, Diversification, Portfolio overlap, Portfolio similarity

1. Introduction

An institutional portfolio can be diversified across many securities yet resemble the portfolios of other institutions. This distinction matters because portfolio theory evaluates concentration within a portfolio (Markowitz, 1952), whereas crowding concerns similarity across portfolios. Common positions, correlated institutional trading, and shared ownership have been associated with prices, comovement, and fragility (Wermers, 1999; Sias, 2004; Coval and Stafford, 2007; Greenwood and Thesmar, 2011; Anton and Polk, 2014).

Existing studies aggregate pairwise holdings overlap among mutual funds (Artiga Gonzalez et al., 2024), show how portfolio-construction practices can generate crowding (Bruno et al., 2018), or use entropy to measure concentration in U.S. equity-market portfolio weights (Perras and Wagner, 2026). We ask a narrower question: are less concentrated 13F portfolios also more distinctive across managers? The analysis uses entropy within each manager and overlap across managers, separating the mechanical role of breadth from the composition of the securities selected.

Using quarterly SEC Form 13F holdings for the 500 largest reported portfolios, we aggregate pairwise minimum-weight overlap into manager-level crowding. The baseline entropy coefficient is 0.067, and a one-standard-deviation difference in the estimation sample corresponds to 0.082 greater crowding. Within the quarterly reselected top-500 sample, mean crowding rises from 0.129 in 2013Q2 to 0.230 in 2025Q4, reflecting both portfolio changes and changes in sample composition.

The contribution is measurement and empirical decomposition. Random breadth accounts for only a limited part of the association, whereas security-popularity and same-quarter peer-consensus diagnostics sharply attenuate it. Exact large-manager exclusions leave the baseline estimate nearly unchanged. The results therefore distinguish within-manager diversification from cross-manager distinctiveness and locate their empirical association in widely shared security choices without identifying why those choices occur.

The design is deliberately descriptive. Following studies that use observed holdings or trading patterns without exogenous variation (Griffin et al., 2011; Ülkü et al., 2023), the regressions document conditional associations rather than the causal effect of diversification. Benchmark replication, liquidity preferences, common mandates or risk models, and correlated flows could jointly determine diversification and overlap. Form 13F supports diagnostics for breadth, widely held securities, institutional consensus, position changes, and large-manager influence, but it does not verify

passive status, benchmarks, mandates, styles, or flows. We therefore treat these exercises as observable-composition diagnostics, not identified mechanisms.

2. Data and methodology

2.1. Sample construction and filing resolution

The sample covers 2013Q2–2025Q4. The SEC structured Form 13F series begins with 2013Q2 holdings, making that quarter the earliest available in the structured-data series and the first quarter in the validated input range (U.S. Securities and Exchange Commission, 2026a). Each quarter, we retain the 500 managers with the largest resolved reported portfolio values. The 51-quarter panel contains 25,500 manager-quarters, 26,637,641 manager-security-quarters, and 628,112 security-quarters. Its pair file contains 6,098,909 nonzero manager-pair-quarter overlaps; zero-overlap pairs remain in every manager-level denominator.

The unit is the reported long-position portfolio in the eligible Section 13(f) security universe, not the institution’s balance sheet. Form 13F omits shorts and incompletely covers non-reportable foreign securities, bonds, and derivatives. The pipeline retains eligible share rows and removes put/call and non-share entries. Entropy and crowding therefore measure observed reportable long positions, not total-firm diversification, leverage, or net exposure.

Values are normalized by accession because the SEC changed 13F reporting from thousands of dollars to dollars on January 3, 2023 (U.S. Securities and Exchange Commission, 2026a, 2022). Summary totals and implied-value diagnostics check earlier same-unit records. Following SEC instructions, a restatement supersedes the earlier filing and an “adds new holdings” amendment supplements it (U.S. Securities and Exchange Commission, 2026b). Unresolved histories are excluded. Of 278,258 source manager-quarter histories, 3,421 (1.23%) are excluded, below the pipeline’s declared 2% integrity gate.

The exploratory return appendix uses a selected 2023Q1–2025Q4 covered subportfolio, legacy stock and SPY adjusted-price caches, a CUSIP-to-symbol crosswalk, and a public Fama–French factor ZIP. Provenance, retrieval records, point-in-time coverage, and redistribution rights for the legacy price files could not be fully verified. The factor ZIP’s public source URL and SHA-256 are recorded. This exercise is exploratory and excluded from the paper’s primary inference.

2.2. Crowding measures and breadth benchmarks

For manager i , security s , and quarter t , let $w_{i,s,t}$ be the security’s portfolio weight. Pairwise overlap is

$$Overlap_{ij,t} = \sum_{s \in S_t} \min(w_{i,s,t}, w_{j,s,t}). \quad (1)$$

This zero-to-one measure follows Artiga Gonzalez et al. (2024). Manager-level crowding averages overlap with all other retained managers, including zero-overlap peers:

$$Crowding_{i,t} = \frac{1}{N_t - 1} \sum_{j \neq i} Overlap_{ij,t}. \quad (2)$$

Diversification is measured by entropy,

$$Entropy_{i,t} = - \sum_s w_{i,s,t} \ln(w_{i,s,t}), \quad (3)$$

where higher values indicate lower concentration. Alternatives are the Herfindahl–Hirschman index, $HHI_{i,t} = \sum_s w_{i,s,t}^2$, and effective holdings, $EffectiveN_{i,t} = 1/HHI_{i,t}$.

The mechanical-breadth benchmark treats two equal-weight portfolios as independent selections of their observed breadths from the quarter’s retained universe. For breadth $n_{i,t}$ and universe size U_t ,

$$E[Overlap_{ij,t} | n_{i,t}, n_{j,t}, U_t] = \frac{\min(n_{i,t}, n_{j,t})}{U_t}. \quad (4)$$

Expected crowding averages this quantity over $N_t - 1$ peers; abnormal crowding is observed minus expected. The benchmark preserves quarter, breadth, and universe, but not weights, popularity, mandates, or benchmark exposures.

A second equal-weight benchmark preserves each manager's holdings counts across up to five tie-preserving, within-quarter categories of securities ranked by number of institutional holders and treats selection as independent within each category. Ties reduce the realized count to four in some quarters. The benchmark retains breadth and coarse popularity exposure, but not portfolio weights or specific securities. For support sets A_i and A_j , Jaccard similarity is $|A_i \cap A_j|/|A_i \cup A_j|$ and binary cosine similarity is $|A_i \cap A_j|/\sqrt{|A_i||A_j|}$. Each support-based alternative is averaged over the same $N_t - 1$ peer set, with zero-overlap peers contributing zero. With 500 managers, the summed overlap in Artiga Gonzalez et al. (2024) equals mean overlap times 499 and only changes scale.

2.3. Regression specification and composition diagnostics

The baseline specification is

$$Crowding_{i,t} = \alpha + \beta Diversification_{i,t} + X'_{i,t}\gamma + \delta_t + \varepsilon_{i,t}, \quad (5)$$

where $X_{i,t}$ includes number of holdings, log portfolio value, and maximum position, and δ_t denotes quarter fixed effects. Standard errors are clustered by manager. To limit sensitivity to boundary portfolios, we exclude manager-quarters with at most five holdings, effective holdings of at least 400, or a maximum position of at least 95%. These rules address, respectively, sparse portfolios, the extreme diversification tail, and near-single-security concentration. They flag 288, 218, and 241 manager-quarters; because the flags overlap, 562 unique observations are excluded, leaving 24,938. Both same- and prior-quarter peer-consensus portfolios are constructed from the full top-500 universe before applying the three regression-sample screens. Because benchmarks, mandates, styles, and flows are unobserved, β is a conditional association rather than a causal effect.

The composition diagnostics are descriptive. Same-quarter consensus overlap compares current weights with the leave-one-out, reported-value-weighted peer portfolio. Prior-quarter alignment instead uses the peer portfolio at $t - 1$, excluding manager i when previously retained; entrants and re-entrants use the full prior-quarter top-500 portfolio. The construction covers 25,000 manager-quarters over 50 transitions, with 94.4% leave-out feasibility and 99.0% mean current-weight coverage. Security and manager exclusions recompute the overlap network. The institutional-footprint split uses within-quarter halves of aggregate retained 13F value. Reported-position change equals one minus consecutive-portfolio overlap. It combines active changes in holdings with passive weight changes caused by relative price movements and is therefore neither a direct measure of trading turnover nor a classification of passive versus active management. We form within-quarter terciles of reported-position change by ranking observations sequentially, breaking ties by ascending manager CIK, and estimate a fully interacted pooled specification, with standard errors clustered by manager. These diagnostics do not identify benchmarks, flows, manager type, investment horizon, or economic mechanisms.

2.4. Identification limits and scope of the diagnostics

To formalize why equation (5) is descriptive, let $Z_{i,t}$ collect latent benchmark exposure, liquidity preference, correlated flows, and common mandate or risk-model constraints:

$$Crowding_{i,t} = \alpha + \beta_Z Diversification_{i,t} + \lambda' Z_{i,t} + X'_{i,t}\gamma + \delta_t + u_{i,t}, \quad (6)$$

$$Diversification_{i,t} = \pi_0 + \pi' Z_{i,t} + X'_{i,t}\rho + \delta_t + v_{i,t}. \quad (7)$$

Here, β_Z denotes the diversification–crowding association conditional on the latent factors $Z_{i,t}$ and need not equal the coefficient β estimated in equation (5). These equations are conceptual and are not estimated. If a component of $Z_{i,t}$ affects both equations, the coefficient in equation (5) may combine the conditional association of interest with omitted-channel dependence.

The empirical diagnostics address restricted alternatives involving the specified mechanical-breadth benchmark, popular-security exposure, and the influence of selected large managers. Popularity adjustment changes the crowding outcome, consensus conditioning adds a holdings-based alignment measure, and security or manager exclusions change the retained universe. These exercises therefore answer different descriptive questions and do not identify separate economic mechanisms. Benchmark tracking, liquidity preference, correlated flows, and shared mandates remain unresolved because benchmark assignments, flows, liquidity, and verified manager styles are unavailable.

3. Empirical results

3.1. Baseline diversification–crowding association

Across 25,500 manager-quarters, mean crowding is 0.159, mean entropy is 4.680, and the average manager reports 1,045 eligible securities (Table A.1). Mean crowding rises from 0.048 in the lowest within-quarter entropy quintile to 0.245 in the highest.

The baseline entropy coefficient is 0.0670 (manager-clustered standard error 0.0030; Table 1). One estimation-sample standard deviation in entropy (1.219) corresponds to 0.0817 greater crowding (manager-clustered 95% confidence interval 0.0744–0.0889), or 50.6% of the regression-sample mean crowding of 0.1613. This is a conditional magnitude, not an intervention effect.

Screening does not drive the result: the entropy coefficient is 0.0648 without the three screens, and estimates obtained by relaxing one screen at a time range from 0.0655 to 0.0668 (Appendix Table A.4).

Effective holdings is positively associated with crowding, whereas HHI is negatively associated; Jaccard and binary-cosine results have the same directional interpretation. Similar current, lagged, and lead coefficients (Appendix Table A.2) do not establish temporal ordering.

Table 1: Diversification and institutional portfolio overlap

Dependent variable or specification	Focal coefficient	Clustered SE	Observations	R^2
Observed crowding: entropy	0.0670	(0.0030)	24,938	0.658
Observed crowding: lagged entropy	0.0651	(0.0030)	23,093	0.654
Observed crowding: effective holdings	0.000255	(0.000049)	24,938	0.503
Observed crowding: HHI	-0.2978	(0.1214)	24,938	0.486
Abnormal crowding: entropy, no holdings-count control	0.0449	(0.0022)	24,938	0.503
Abnormal crowding: entropy, holdings-count controlled	0.0531	(0.0030)	24,938	0.513
Jaccard similarity	0.0639	(0.0023)	24,938	0.707
Binary cosine similarity	0.0912	(0.0034)	24,938	0.714
Summed minimum-weight overlap ($499 \times$ mean)	33.4206	(1.5062)	24,938	0.658
Within-manager Δ abnormal crowding	0.0260	(0.0033)	23,044	0.247

Notes: Each row reports a separate regression with quarter fixed effects, manager-clustered standard errors, and controls for log portfolio value, maximum position, and, except where indicated, holdings count. Abnormal crowding subtracts the random-breadth benchmark; summed overlap is a 499-fold scale translation of mean crowding; and the first-difference specification uses changes in the control variables.

3.2. Composition adjustments and position-change heterogeneity

The random-breadth adjustment attenuates but does not eliminate the association: entropy coefficients are 0.04493 without and 0.05309 with holdings count controlled. The controlled within-manager first-difference coefficient is 0.02596 (Table 1). These results indicate that the levels association is not solely the arithmetic consequence of different breadths, although the positive lead diagnostic still prevents temporal identification.

The tie-preserving popularity benchmark is more consequential: its entropy coefficient is 0.00176 without and 0.00920 with holdings count controlled (Table 2). Observed crowding remains highly correlated with random-breadth abnormal crowding, Jaccard similarity, binary cosine similarity, and same-quarter consensus (Pearson/Spearman coefficients of 0.8685–0.9841), but less strongly with popularity-adjusted abnormal crowding (0.6473/0.6595; Appendix Table A.8). These same-input concordances are descriptive, not independent validation.

Conditioning on same-quarter consensus changes the entropy coefficient to -0.00561 . The prior-quarter peer-alignment diagnostic defined in Section 2.3 produces a similar result: on the 24,447-observation common sample, the coefficient changes from 0.06711 to -0.00615 , while alignment enters at 0.44920 (Table 2). Its raw correlation with current crowding is 0.9727. These negative conditional slopes are residual compositional associations, not causal effects or identified channels.

Almost all measured overlap is concentrated in high-institutional-footprint securities. Their mean contribution and entropy coefficient are 0.15934 and 0.06697, versus 0.0000047 and 0.0000004 for low-footprint securities (Table 2). This is an additive retained-13F-value split, not a market-capitalization or liquidity split.

The reported-position-change terciles group observations by weight changes, which include price drift; they do not classify passive/index institutions and active managers. In the fully interacted model, entropy slopes decline from 0.08899 to 0.07323 and 0.05047 across the low-, middle-, and high-change terciles. Equality is rejected ($\chi^2(2) = 52.58$, $p < 0.001$), and all pairwise differences have $p \leq 0.0011$. Because reported weight change includes price drift, this heterogeneity does not identify turnover, manager type, or horizon. Relative to the baseline coefficient of 0.06698, removing the 50 or 100 most popular securities yields 0.05209 and 0.04431; excluding the name-matched Big Three or top 10 managers yields 0.06595 and 0.06543 (Appendix Table A.4).

Table 2: Observable-composition and reported-position-change diagnostics

Dependent variable, sample, or test	Estimate or test	Clustered SE	Observations	R^2
<i>Panel A: Observable composition</i>				
Popularity-adjusted, no holdings-count control	0.0018	(0.0019)	24,938	0.136
Popularity-adjusted, holdings-count controlled	0.0092	(0.0032)	24,938	0.152
Entropy conditional on consensus overlap	-0.0056	(0.0009)	24,938	0.974
Consensus overlap in joint specification	0.4411	(0.0046)	24,938	0.974
Prior-consensus common-sample baseline entropy	0.0671	(0.0030)	24,447	0.658
Entropy conditional on prior-quarter peer alignment	-0.0061	(0.0009)	24,447	0.974
Prior-quarter peer alignment in joint specification	0.4492	(0.0046)	24,447	0.974
High institutional-footprint contribution	0.0670	(0.0030)	24,938	0.658
Low institutional-footprint contribution	0.0000004	(0.0000004)	24,938	0.238
<i>Panel B: Fully interacted reported-position change</i>				
Low-change entropy slope	0.0890	(0.0041)	23,093	0.697
Middle-change entropy slope	0.0732	(0.0043)	23,093	0.697
High-change entropy slope	0.0505	(0.0038)	23,093	0.697
Middle minus low entropy slope	-0.0158	(0.0048)	23,093	0.697
High minus low entropy slope	-0.0385	(0.0054)	23,093	0.697
High minus middle entropy slope	-0.0228	(0.0047)	23,093	0.697
Equal entropy slopes	$\chi^2(2) = 52.58$; $p < 0.001$		–	23,093 0.697

Notes: Panel A includes quarter effects and portfolio controls (holdings count omitted in the first row). Panel B is fully interacted across within-quarter reported-position-change terciles ($N = 23,093$; 1,008 manager clusters). Standard errors are manager-clustered. Reported-position change combines active changes in holdings with passive price-driven weight changes; the terciles do not classify institutions as passive/index or active and do not measure trading turnover.

3.3. Crowding trend and aggregate correlates

Within the quarterly reselected top-500 sample, average crowding rises from 0.129 in 2013Q2 to 0.230 in 2025Q4 (Figure 1). This aggregate increase combines portfolio changes among continuing managers with entry and exit from the selected sample; it does not establish an increase for a fixed set of institutions. Random-breadth expected crowding rises from 0.022 to 0.041, while abnormal crowding rises from 0.108 to 0.189.

Of the 0.100 increase in observed crowding, 0.019 appears in expected random-breadth crowding and 0.081 in abnormal crowding. The fitted trend is 0.0064 per year; the series is not monotonic and declines in 2022.

First-difference correlations of crowding with average entropy, average holdings, and top-10 and top-50 institutional-footprint shares are 0.435, 0.712, 0.726, and 0.668; detrended correlations are 0.530, 0.627, 0.761, and 0.814 (Appendix Table A.3). The footprint shares are fractions of aggregate retained 13F holding value. Changes in breadth and top-security institutional-value concentration covary with crowding, but these diagnostics do not identify passive growth or another structural cause. The observed-versus-random-breadth decomposition does not separate changes among continuing managers from changes in sample composition.

3.4. Interpretation and identification limits

Entropy and crowding capture different dimensions of portfolio structure: entropy describes weight concentration within a manager, whereas crowding describes similarity across managers. The positive coefficients under the

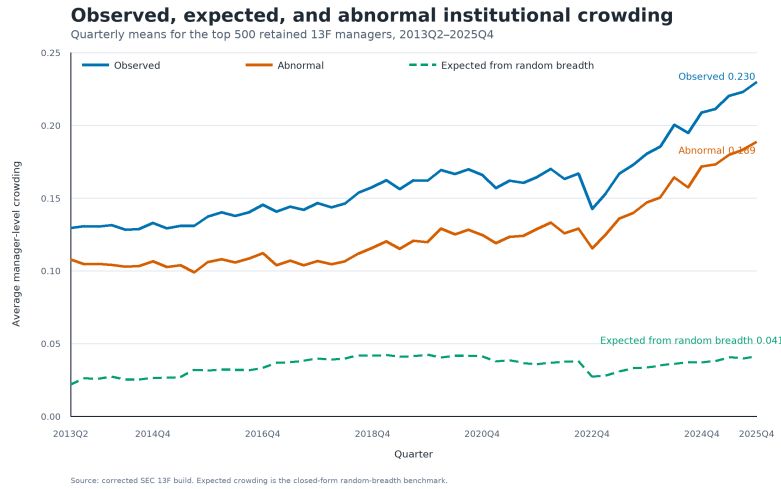


Figure 1: Quarterly mean observed, random-breadth expected, and abnormal crowding, 2013Q2–2025Q4, in the quarterly reselected top-500 sample. Abnormal crowding equals observed crowding minus expected overlap under independent equal-weight selection at managers’ observed breadths. Changes in the plotted means reflect both portfolio changes and changes in sample composition.

specified breadth adjustment and the reported popular-security and large-manager exclusions show that those particular adjustments do not eliminate the association. Popularity-adjusted outcomes and consensus-conditioned regressions instead show substantial dependence on shared holdings composition. Because these exercises change the outcome, conditioning variables, or retained universe, their estimates are not a causal decomposition of the baseline association. They do not distinguish benchmark tracking, liquidity preference, correlated flows, or shared mandates. The results therefore inform the measurement of institutional crowding, not the causal effects of diversification or claims about systemic fragility. Appendix A.1 reports complementary exploratory diagnostics for the selected price-covered subportfolio. Selection is substantial: retained manager-quarters have mean crowding of 0.231, compared with 0.135 among excluded observations (Appendix Table A.5). Within this selected sample, crowding is positively associated with peer-return correlation and downside peer beta, but standalone downside outcomes are not consistently positive. These exploratory associations cannot be generalized to the full holdings sample, and coverage-threshold sensitivity does not by itself resolve the selection problem.

4. Conclusion

Filing-resolved SEC Form 13F holdings show that managers with less concentrated reported long portfolios also overlap more with their peers. The association remains positive after random-breadth adjustment, alternative similarity measures, within-manager changes, and exact manager exclusions, but security-popularity and same-quarter peer-consensus diagnostics sharply attenuate it. These results locate the empirical pattern in widely shared security choices without identifying why managers make those choices.

Within the quarterly reselected top-500 sample, mean crowding rises from 2013Q2 through 2025Q4 by substantially more than the random-breadth benchmark. This aggregate trend combines portfolio changes among continuing managers with entry and exit from the selected sample; it does not establish a within-manager increase for a fixed cohort.

The findings pertain to eligible reported Section 13(f) long-share positions, not institutions’ complete portfolios, balance sheets, or net exposures. They do not identify causal channels or establish systemic-risk effects. They document a narrower distinction: within-portfolio diversification and cross-institution distinctiveness are separate dimensions, so peer similarity should be measured directly rather than inferred from portfolio concentration.

Data availability

The study uses public SEC 13F structured data. The legacy stock and SPY adjusted-price caches and CUSIP-to-symbol crosswalk used only in the exploratory appendix are not redistributed because their provenance and point-in-time coverage could not be fully verified. The public Fama–French factor ZIP has a recorded source URL and SHA-256.

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Appendix A. Additional descriptive and diagnostic evidence

Table A.1 reports manager-quarter summaries and within-quarter entropy quintiles. Table A.2 compares sample windows and timing diagnostics. Table A.3 reports time-series decompositions. Appendix Table A.4 reports additional sample-composition and screening robustness tests. None creates exogenous variation.

Table A.1: Summary statistics and entropy quintiles

<i>Panel A: Manager-quarter summary statistics</i>						
Variable	Mean	SD	P25	Median	P75	N
Crowding	0.1593	0.1000	0.0735	0.1548	0.2422	25,500
Entropy	4.680	1.328	3.867	4.936	5.716	25,500
HHI	0.0464	0.1220	0.0075	0.0143	0.0333	25,500
Effective N	93.8	86.5	30.0	70.1	132.7	25,500
Number of holdings	1,044.6	1,274.7	129.0	594.5	1,437.0	25,500
Log portfolio value	23.627	1.176	22.759	23.318	24.218	25,500
Maximum position	0.0917	0.1408	0.0313	0.0503	0.0844	25,500
<i>Panel B: Within-quarter entropy quintiles</i>						
Entropy quintile	Mean crowding	Mean entropy	Mean holdings	Mean effective N	Mean HHI	N
Quintile 1	0.0479	2.621	83.3	13.6	0.1724	5,100
Quintile 2	0.1056	4.091	338.5	40.1	0.0294	5,100
Quintile 3	0.1672	4.926	689.9	78.1	0.0150	5,100
Quintile 4	0.2305	5.563	1,329.7	123.3	0.0094	5,100
Quintile 5	0.2455	6.197	2,781.7	213.8	0.0056	5,100

Notes: Panel A uses all 25,500 manager-quarters. Panel B sorts managers separately within each quarter and reports 5,100 observations per quintile.

Table A.2: Subsample and lead-lag diagnostics

Sample or timing specification	Entropy coefficient	Clustered SE	Observations	R^2
Extended sample, 2013Q2–2025Q4	0.0670	(0.0030)	24,938	0.658
Early subsample, 2013Q2–2017Q4	0.0651	(0.0032)	9,292	0.704
Recent subsample, 2018Q1–2025Q4	0.0689	(0.0035)	15,646	0.619
Current entropy	0.0449	(0.0022)	24,938	0.503
Lagged entropy	0.0440	(0.0023)	23,093	0.498
Lead-entropy placebo	0.0439	(0.0022)	23,044	0.494

Notes: The first three rows use observed crowding and the baseline controls. The timing rows use abnormal crowding and control for log portfolio value and maximum position, but not number of holdings. The similar current, lagged, and lead coefficients indicate persistence and do not establish temporal identification.

Table A.3: Time-trend decomposition and descriptive correlates

Series	2013Q2	2025Q4	Change	First-difference corr.	Detrended corr.
Observed crowding	0.129	0.230	0.100	–	–
Expected random-breadth crowding	0.022	0.041	0.019	–	–
Abnormal crowding	0.108	0.189	0.081	–	–
Average entropy	–	–	–	0.435	0.530
Average holdings	–	–	–	0.712	0.627
Top-10 institutional footprint	–	–	–	0.726	0.761
Top-50 institutional footprint	–	–	–	0.668	0.814

Notes: Observed crowding's fitted linear change is 0.0064 per year. Correlations use quarterly mean crowding and the named quarterly series. The top-500 sample is reselected each quarter. The observed-versus-random-breadth decomposition does not separate within-manager change from manager entry and exit and does not provide structural attribution.

Table A.4: Popular-security, large-manager, and portfolio-screening sensitivity

Specification	Entropy coefficient	Clustered SE	Observations	R ²
<i>Panel A: Popular-security exclusions</i>				
Remove top 50 popular securities	0.0521	(0.0018)	23,971	0.767
Remove top 100 popular securities	0.0443	(0.0015)	23,567	0.784
<i>Panel B: Large-manager exclusions</i>				
All managers (baseline three screens)	0.0670	(0.0030)	24,938	0.658
Exclude 11 name-matched Big Three CIKs	0.0660	(0.0030)	24,705	0.653
Exclude top 3 managers each quarter	0.0665	(0.0030)	24,785	0.653
Exclude top 5 managers each quarter	0.0662	(0.0030)	24,683	0.651
Exclude top 10 managers each quarter	0.0654	(0.0030)	24,439	0.646
<i>Panel C: Portfolio-screening sensitivity</i>				
Baseline three screens	0.0670	(0.0030)	24,938	0.658
No portfolio screens	0.0648	(0.0030)	25,500	0.658
Relax holdings-count screen	0.0668	(0.0030)	25,041	0.661
Relax effective-holdings screen	0.0655	(0.0030)	25,156	0.647
Relax maximum-position screen	0.0667	(0.0030)	24,994	0.660

Notes: Panels A and B retain the baseline three portfolio screens and recompute portfolio metrics and the overlap network after security or manager exclusions. Panel C changes only the regression-sample screens; each one-screen-relaxed specification retains the other two screens, while crowding remains constructed from the full quarterly top-500 universe. Manager ranks are quarter specific, and the name-matched Big Three test does not verify affiliation.

Appendix A.1. Exploratory price-covered common-movement and downside diagnostics

Appendix A.1 reports exploratory next-quarter return diagnostics for the price-covered subportfolio described in Section 2.1. Portfolio weights measured at quarter-end are applied only to subsequent-quarter returns, preserving the temporal ordering between holdings information and realized outcomes. Recent financial-prediction studies have applied deep-learning methods to global stock-index returns and emphasized carefully aligning predictive information with subsequent market outcomes (Hu and Shen, 2026a,b). Table A.5 summarizes price-coverage selection and sensitivity: retained manager-quarters have mean crowding of 0.231, compared with 0.135 among excluded observations. The estimates therefore concern a selected group with higher crowding, and varying the coverage gate does not by itself correct this selection. Tables A.6 and A.7 report common-movement, downside, and factor-residual results.

Table A.6 reports positive associations with peer correlation and downside peer beta, while other standalone downside outcomes are imprecise. Residual-beta estimates are mixed (Table A.7), so the evidence indicates selected-sample downside comovement but no consistent increase in standalone downside loss.

Table A.5: Selection balance and price-coverage sensitivity

Panel A: Price-coverage selection balance							
Sample	Manager-quarters		Mean crowding	Mean entropy	Mean holdings	Mean priced weight	
Retained	3,780		0.231	4.618	1,128.2	0.722	
Excluded	2,220		0.135	4.676	1,307.4	0.284	
Panel B: Coverage-gate sensitivity							
Minimum priced weight	Manager-quarters	Minimum managers/quarter		Coefficient	Manager SE	Quarter SE	Observations
50%	3,780		298	0.7246	(0.0676)	(0.0846)	3,734
60%	3,097		222	0.7719	(0.0839)	(0.0996)	3,053
70%	2,095		128	0.8373	(0.1526)	(0.1227)	2,054
80%	1,058		49	0.6766	(0.1333)	(0.1074)	1,017

Notes: Panel A compares all 6,000 manager-quarters before the primary 50% coverage gate. Panel B rebuilds the leave-one-out peer return at each gate and reports the observed-crowding coefficient from the same controlled specification. The selected sample has higher mean crowding than excluded observations. Varying the coverage gate does not by itself correct selection bias; these selected-subportfolio estimates cannot be generalized to the full holdings sample and are not causal estimates.

Table A.6: Exploratory next-quarter common-movement and downside diagnostics

Outcome and focal variable	Coefficient	Manager-clustered SE	Quarter-clustered SE	Observations
Own-peer return correlation: observed crowding	0.7246	(0.0676)	(0.0846)	3,734
Own-peer return correlation: abnormal crowding	0.7394	(0.0679)	(0.0851)	3,734
Fisher-z own-peer correlation: observed crowding	6.3494	(0.2432)	(0.1603)	3,734
Raw peer beta: observed crowding	0.6110	(0.1958)	(0.0907)	3,734
Downside peer beta: observed crowding	0.6837	(0.2185)	(0.1309)	3,734
Realized volatility: observed crowding	-0.0220	(0.0274)	(0.0232)	3,734
Downside volatility: observed crowding	-0.0184	(0.0195)	(0.0170)	3,734
Maximum drawdown: observed crowding	0.0238	(0.0132)	(0.0183)	3,734
SPY-adjusted negative-return accumulation	0.0665	(0.0385)	(0.0389)	3,734

Notes: Regressions use quarter fixed effects and baseline controls on the 2023Q1–2025Q4 price-covered sample. Manager-clustered standard errors are primary; quarter-clustered errors are diagnostic given 12 quarters. Covered weights are renormalized after the 50% coverage gate and applied to next-quarter returns. Fisher-z is used for the bounded correlation outcome.

Table A.7: Factor-residual common-movement diagnostics

Outcome	Crowding coefficient	Manager-clustered SE	Quarter-clustered SE	Observations
SPY-residual correlation	1.9549	(0.1513)	(0.3196)	3,734
SPY-residual peer beta	0.7333	(0.7781)	(1.0632)	3,734
SPY-residual downside beta	1.1446	(0.7598)	(1.5144)	3,734
FF3-residual correlation	1.0921	(0.1393)	(0.2108)	3,734
FF3-residual peer beta	-1.5481	(0.5928)	(0.6505)	3,734
FF3-residual downside beta	-1.2751	(0.7155)	(0.8380)	3,734

Notes: All rows use observed crowding. SPY and Fama–French-three-factor residual correlations remain positive, but residual beta estimates are mixed. The same 12-quarter, price-covered, renormalized-subportfolio limitations as Table A.6 apply.

Appendix A.2. Crowding-measure correlation matrix

Table A.8 reports cross-measure correlations on the common screened holdings sample. It is a descriptive alignment check, not independent validation or a mechanism test.

Table A.8: Pearson and Spearman correlations among crowding measures and diagnostics

Measure	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pearson correlations</i>						
(1) Observed crowding	1.0000	–	–	–	–	–
(2) Random-breadth abnormal	0.9739	1.0000	–	–	–	–
(3) Popularity-adjusted abnormal	0.6473	0.7800	1.0000	–	–	–
(4) Jaccard similarity	0.8685	0.7798	0.2444	1.0000	–	–
(5) Binary-cosine similarity	0.8759	0.7906	0.2791	0.9920	1.0000	–
(6) Leave-one-out consensus	0.9728	0.9229	0.5578	0.8690	0.8681	1.0000
<i>Panel B: Spearman rank correlations</i>						
(1) Observed crowding	1.0000	–	–	–	–	–
(2) Random-breadth abnormal	0.9761	1.0000	–	–	–	–
(3) Popularity-adjusted abnormal	0.6595	0.7788	1.0000	–	–	–
(4) Jaccard similarity	0.8715	0.7892	0.2882	1.0000	–	–
(5) Binary-cosine similarity	0.8831	0.8013	0.3153	0.9958	1.0000	–
(6) Leave-one-out consensus	0.9841	0.9402	0.5813	0.8871	0.8949	1.0000

Notes: Both panels use the same common screened sample of 24,938 manager-quarters. Panel A reports Pearson correlations and Panel B reports Spearman rank correlations. Observed and random-breadth abnormal crowding use minimum-weight overlap; the popularity-adjusted measure preserves coarse security-popularity exposure; Jaccard and binary cosine use support sets. Leave-one-out consensus is computed from the same reported holdings. The matrices document same-input concordance; they are neither an independent external validation nor evidence of an economic mechanism.